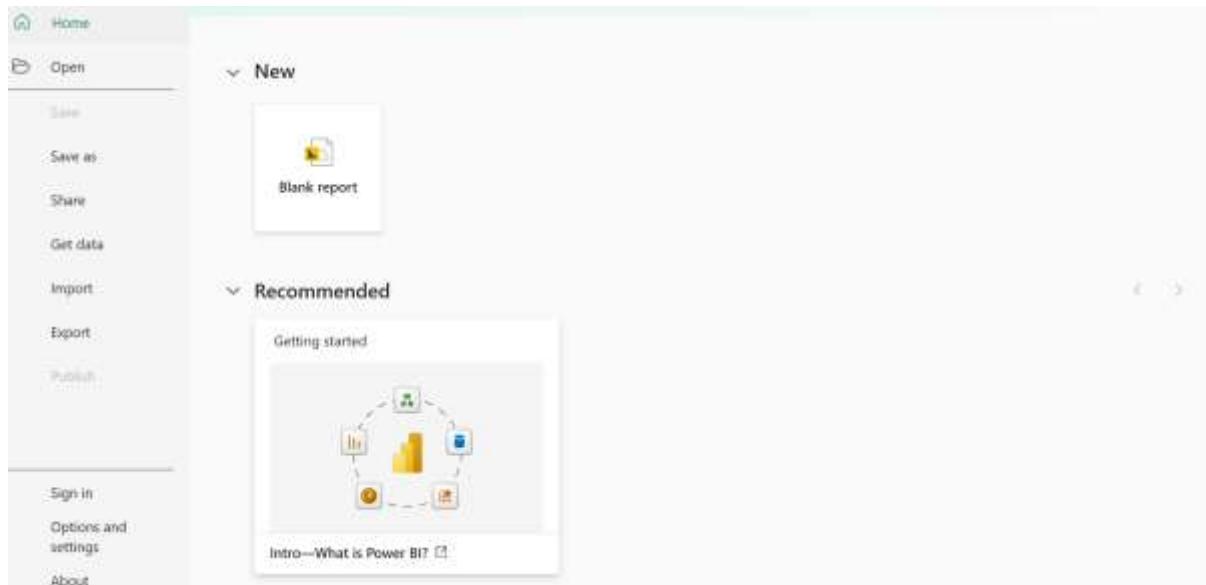


WEEK 1 – POWER BI REPORT

Power BI Desktop Home Page

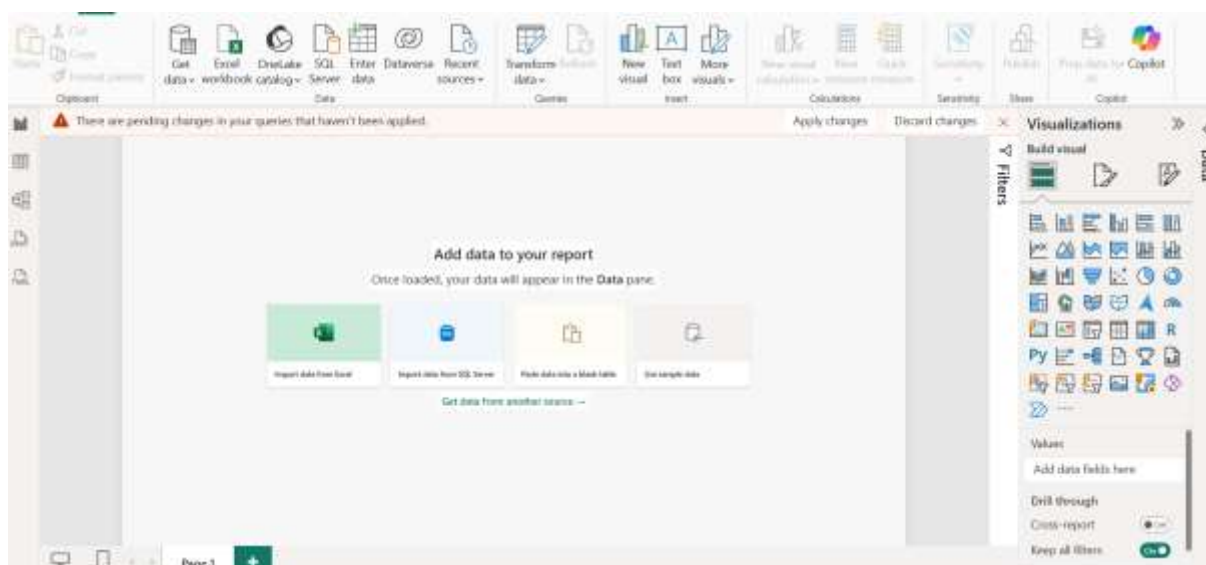
Power BI Desktop provides a simple interface to create and manage reports. The Home page includes options such as Blank Report, Get Data, and Import. The Recommended section provides useful resources for beginners. This interface serves as the starting point for creating interactive Power BI reports.



Power BI Desktop Home Page with Report Creation and Learning Options

Power BI Desktop Interface

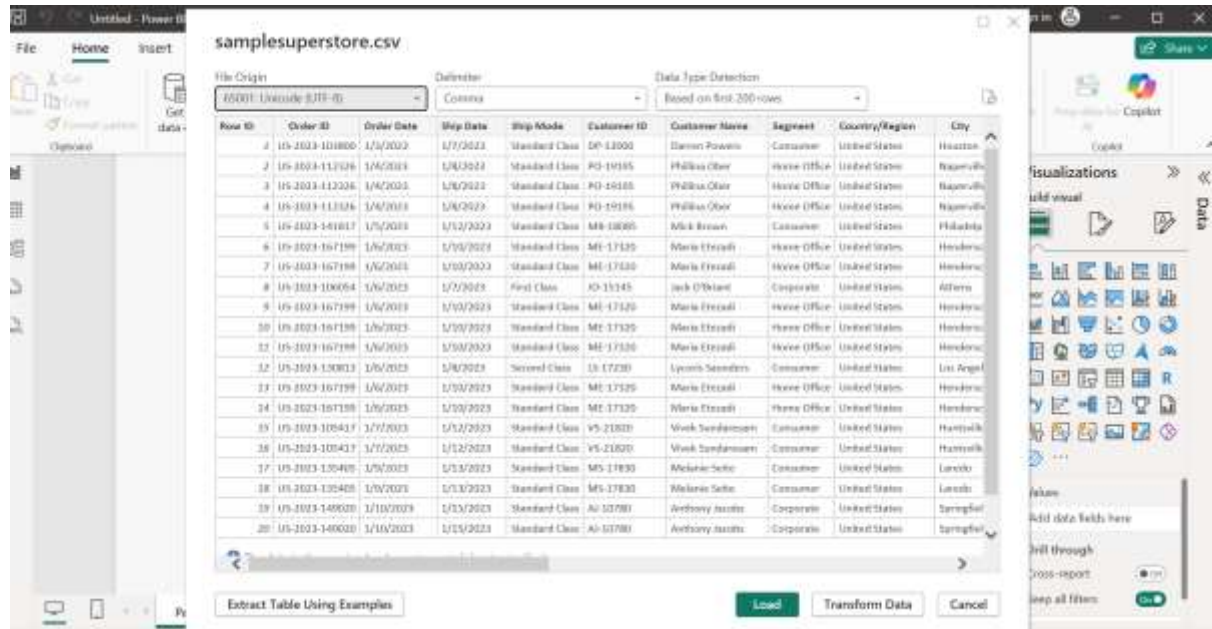
Power BI Desktop was opened to begin the Superstore Sales data analysis process. The main workspace provides access to essential features such as data import, transformation, modeling, and report creation. This interface serves as the starting point for preparing the dataset and developing the Power BI report.



Power BI Desktop workspace used to initiate the data analysis and reporting process

Importing the Superstore Sales Dataset

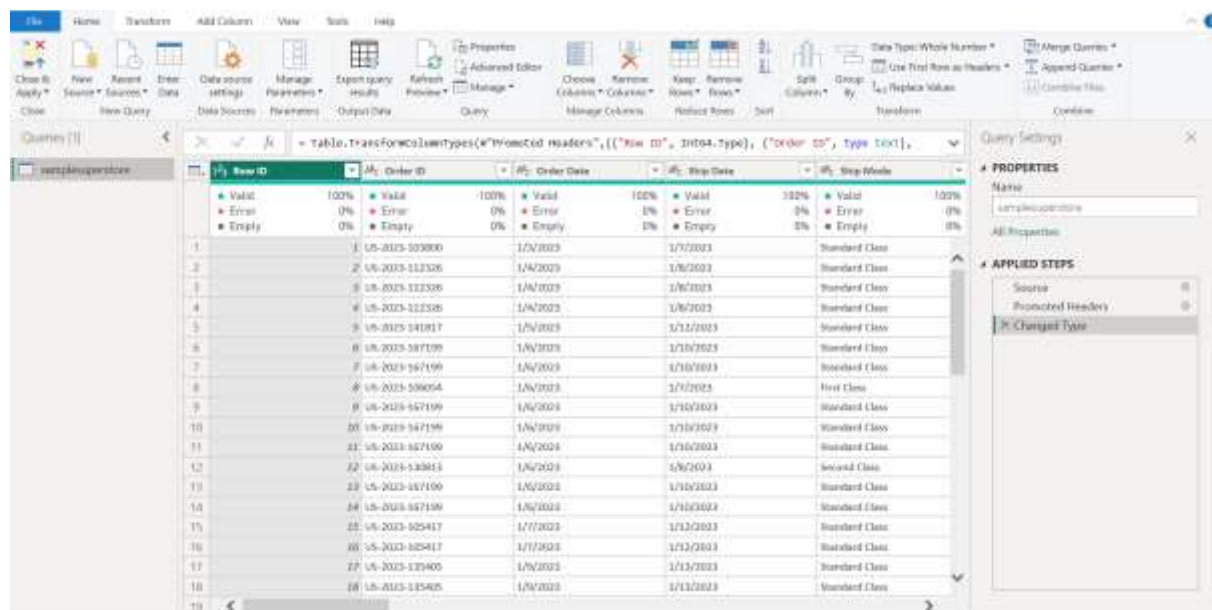
The Superstore Sales CSV file was selected for import into Power BI. The preview window displays the available fields and sample records, allowing the dataset structure to be checked before loading. The file can then be loaded directly or opened in Power Query for further data preparation.



Preview of the Superstore Sales CSV file before importing the data into Power BI

Reviewing the Dataset in Power Query

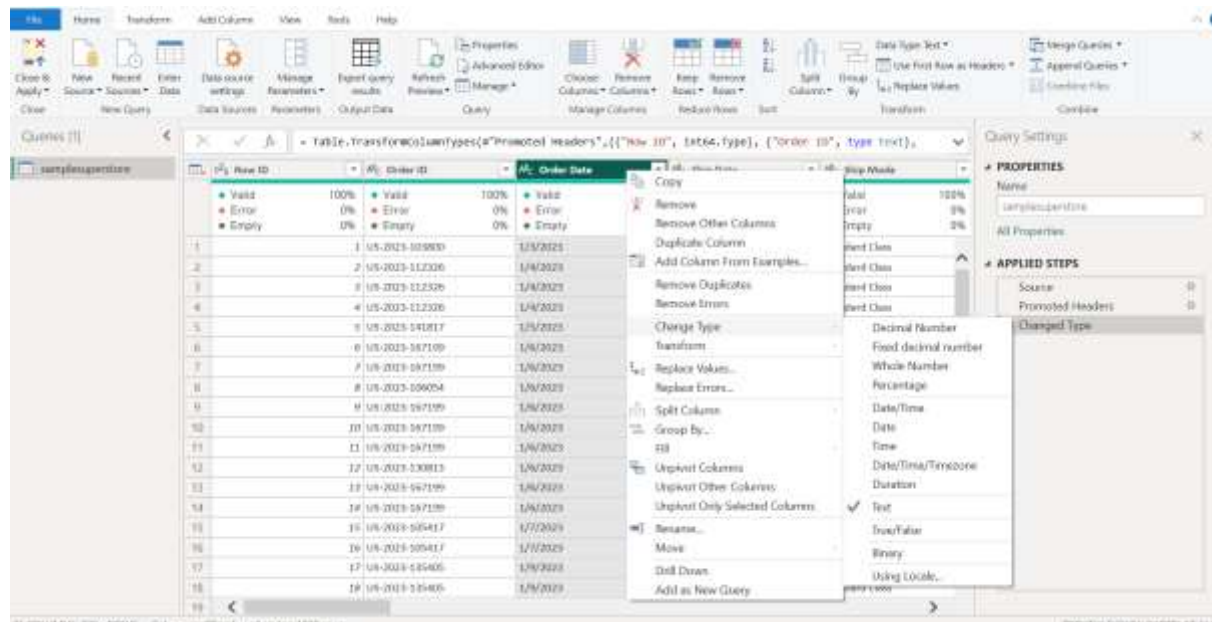
The Superstore dataset was opened in Power Query Editor to examine its structure and existing fields. The displayed columns were reviewed along with their data types and data quality indicators. The Applied Steps section also shows the initial actions performed on the imported dataset, providing a clear starting point for the upcoming data transformation process.



Superstore dataset reviewed in Power Query Editor before applying further transformations

Power Query Editor – Data Transformation

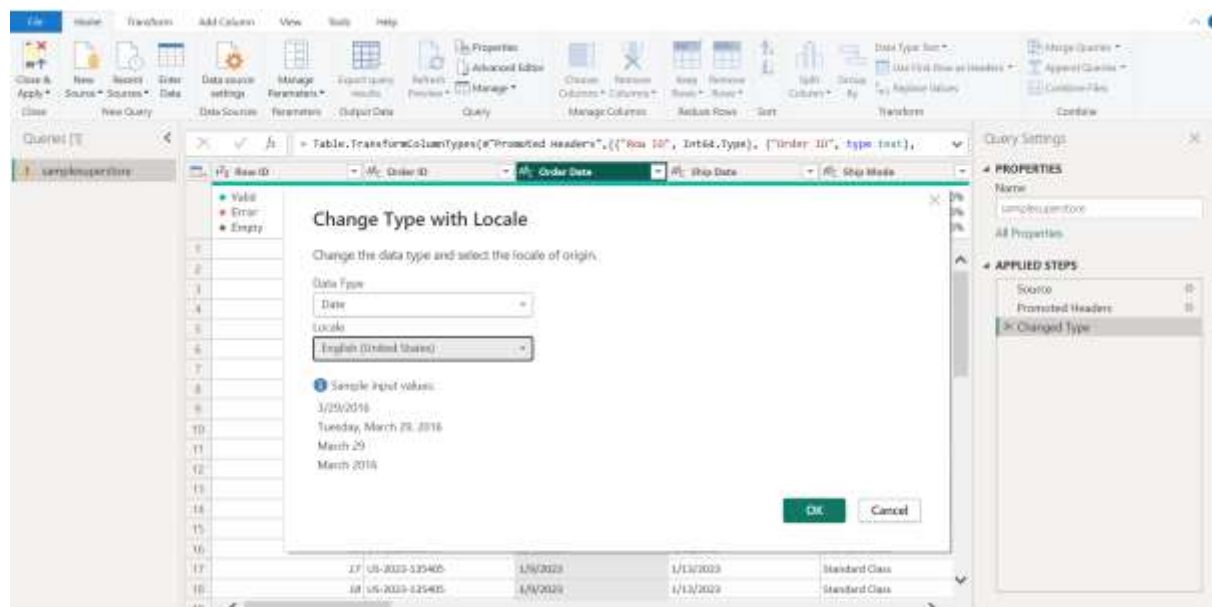
The Power Query Editor is used to clean and transform data before loading it into Power BI. It provides options to remove duplicates, errors, and unnecessary columns. Users can change the data type of columns, such as Text, Date, and Number. The Applied Steps pane records all transformations performed on the dataset.



Power Query Editor Showing Data Transformation and Data Type Options

Converting Order Date with Locale

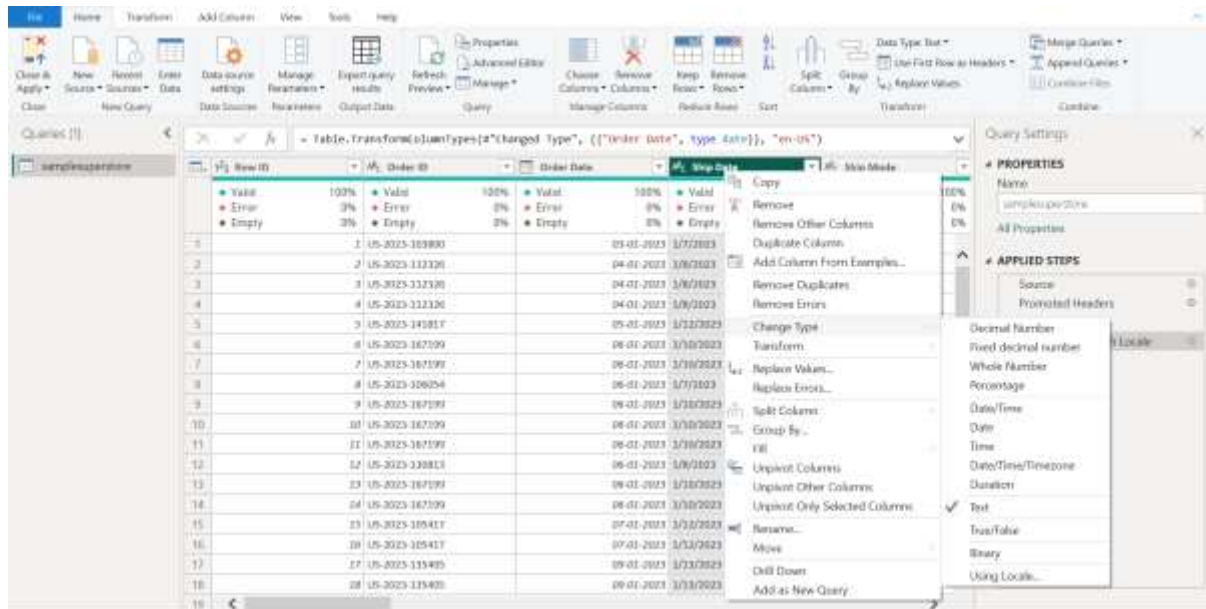
The Order Date column was configured with the Date data type in Power Query Editor. The English (United States) locale was selected to ensure that the date values are interpreted in the appropriate format. This step helps maintain consistency and accuracy while working with date-related data.



Configuring the Order Date column as Date type using the English (United States) locale

Changing Data Types in Power Query

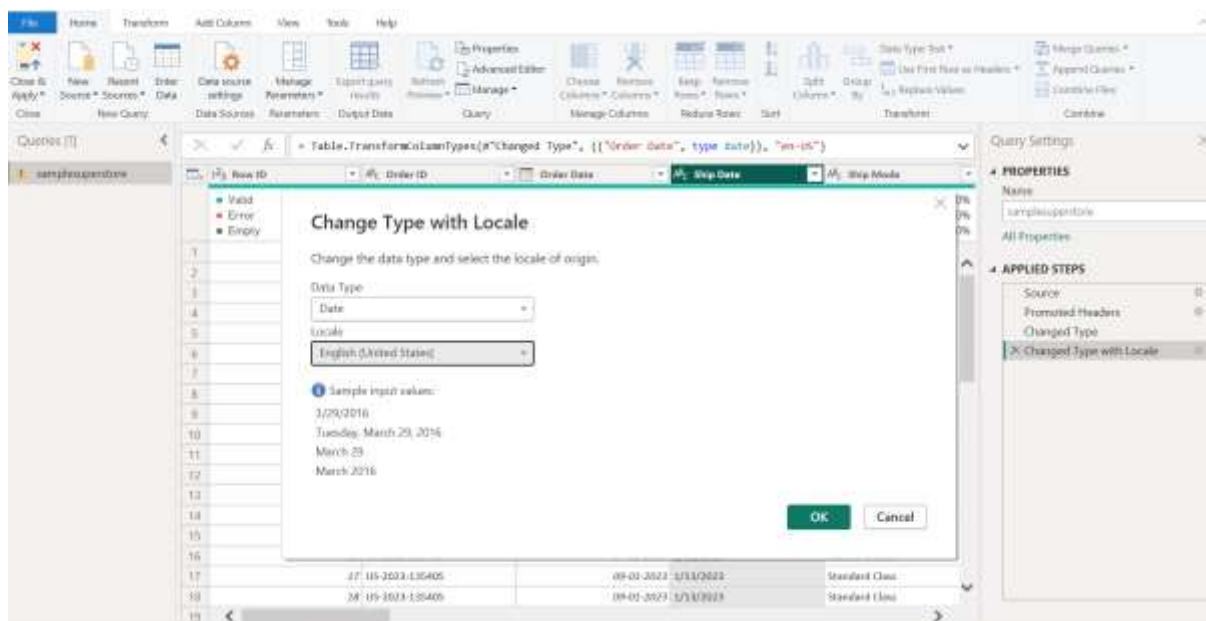
The Change Type option allows users to assign the correct data type to each column. In this example, the Order Date column is converted into a Date format. Power Query provides different types such as Text, Whole Number, Decimal, Date, and Date/Time. Selecting the correct data type ensures accurate data processing and analysis.



Changing the Order Date Column to the Appropriate Data Type

Setting the Date Format and Locale

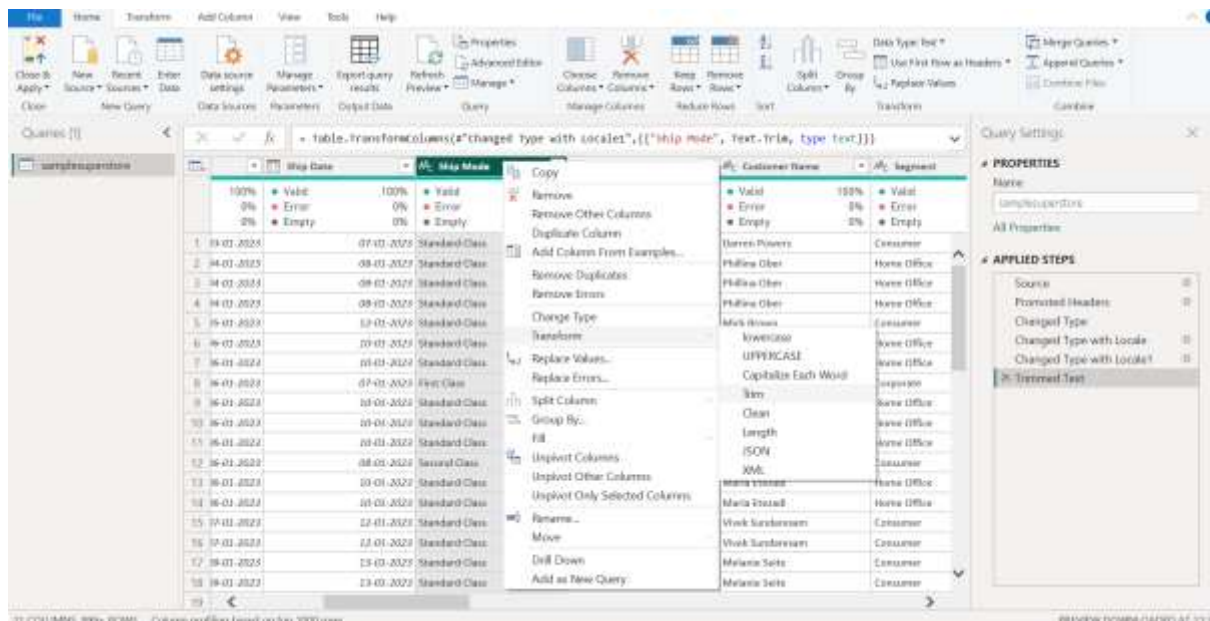
The selected field was configured with the Date data type through the Change Type with Locale option. The English (United States) locale was chosen to ensure that the date values are interpreted correctly. This configuration helps maintain a consistent format for the date field during the data preparation process.



Selecting the Date data type and locale for accurate date interpretation

Text Trimming in Power Query

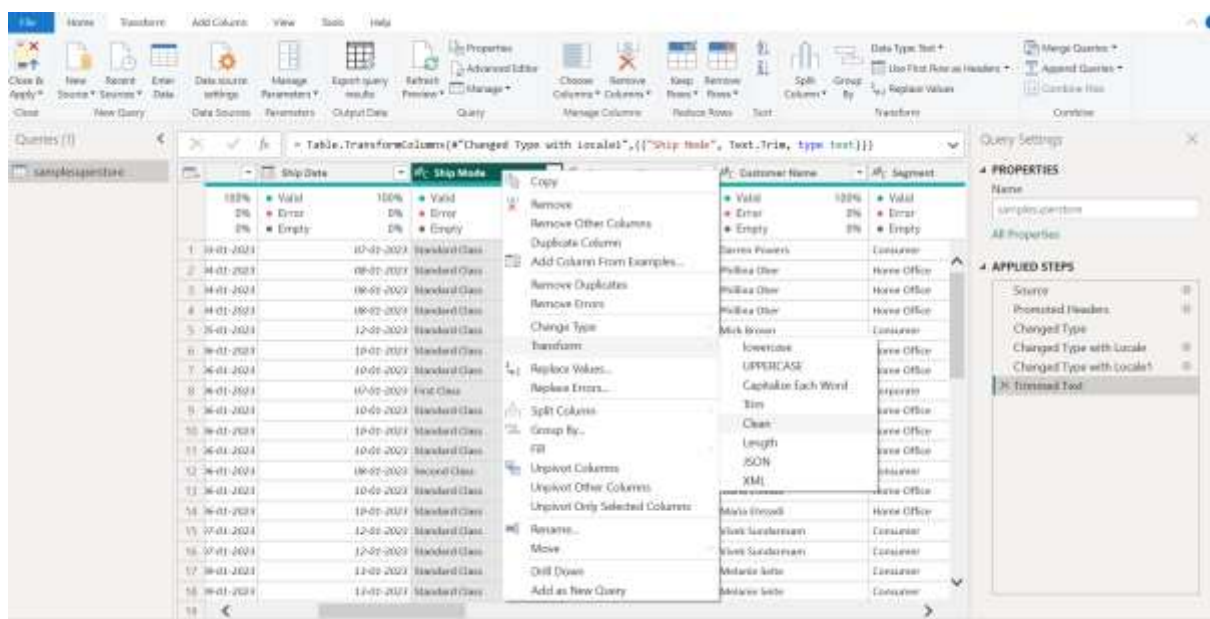
The Transform option provides useful tools for cleaning and formatting text data. In this example, the Ship Mode column is being processed using text transformation options. Power Query allows users to apply functions such as Trim, Clean, Uppercase, Lowercase, and Capitalize Each Word. These transformations help maintain consistent and accurate text values for further analysis.



Applying Text Transformation Options in Power Query

Text Cleaning and Formatting in Power Query

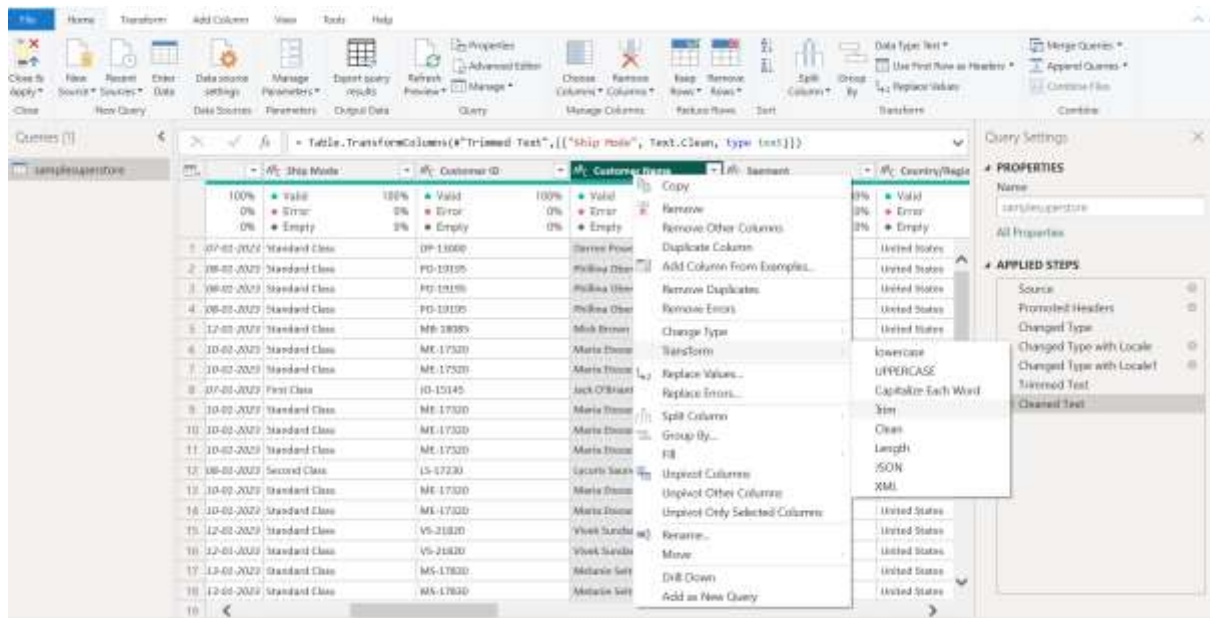
The Transform menu provides several options to clean and standardize text values. In this example, the Ship Mode column is selected for text transformation. Options such as Trim, Clean, Uppercase, Lowercase, and Capitalize Each Word help improve data consistency. These transformations prepare the text data for accurate analysis and reporting.



Using Text Transformation and Cleaning Options in Power Query

Trimming Text Data in Power Query

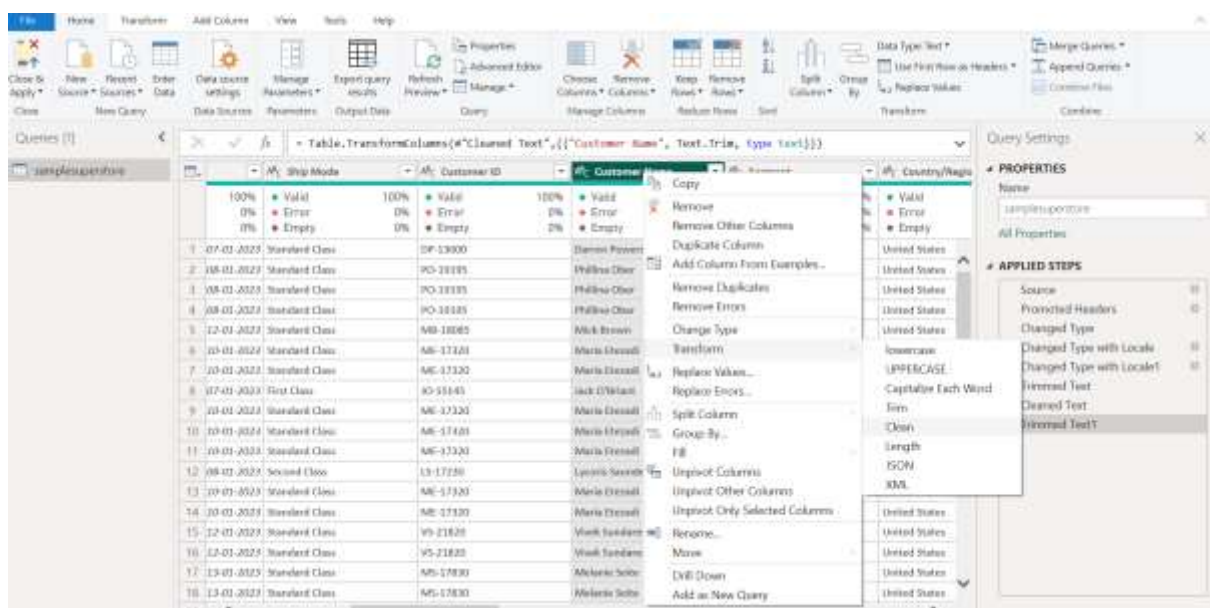
The Trim transformation removes unnecessary spaces from text values. In this example, the Customer Name column is selected for Cleaning. This helps ensure that names are stored without extra spaces before or after the text. Cleaning and trimming the data improves consistency and accuracy during analysis.



Cleaning and Formatting Customer Name Data in Power Query

Cleaning Text Data in Power Query

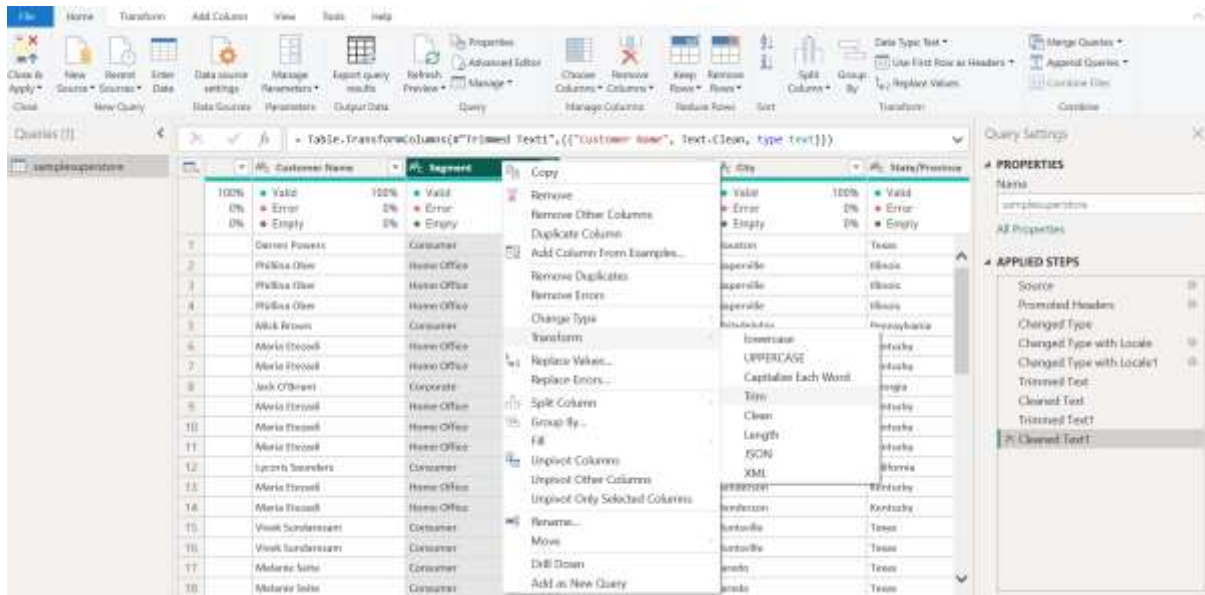
The Clean transformation helps remove unwanted or non-printable characters from text values. In this example, the Customer Name column is selected for text cleaning. Power Query also provides options such as Trim, Uppercase, Lowercase, and Capitalize Each Word. These functions help improve the quality and consistency of customer information.



Applying the Trim Transformation to Customer Name Data

Cleaning Segment Data in Power Query

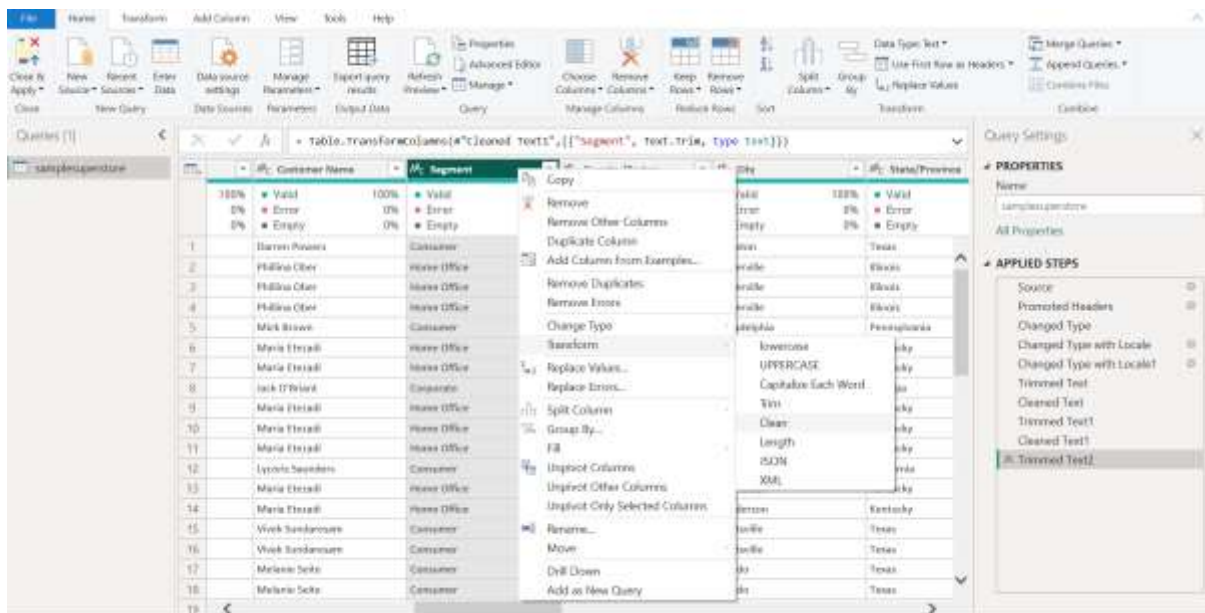
The Transform menu provides options to clean and standardize segment information. In this example, the Customer Name column is selected for text transformation. The Clean functions help remove unwanted spaces and characters from the data. This ensures that segment names are consistent, clean, and ready for accurate analysis.



Applying Clean Transformations to Segment Name Data

Trimming Segment Data in Power Query

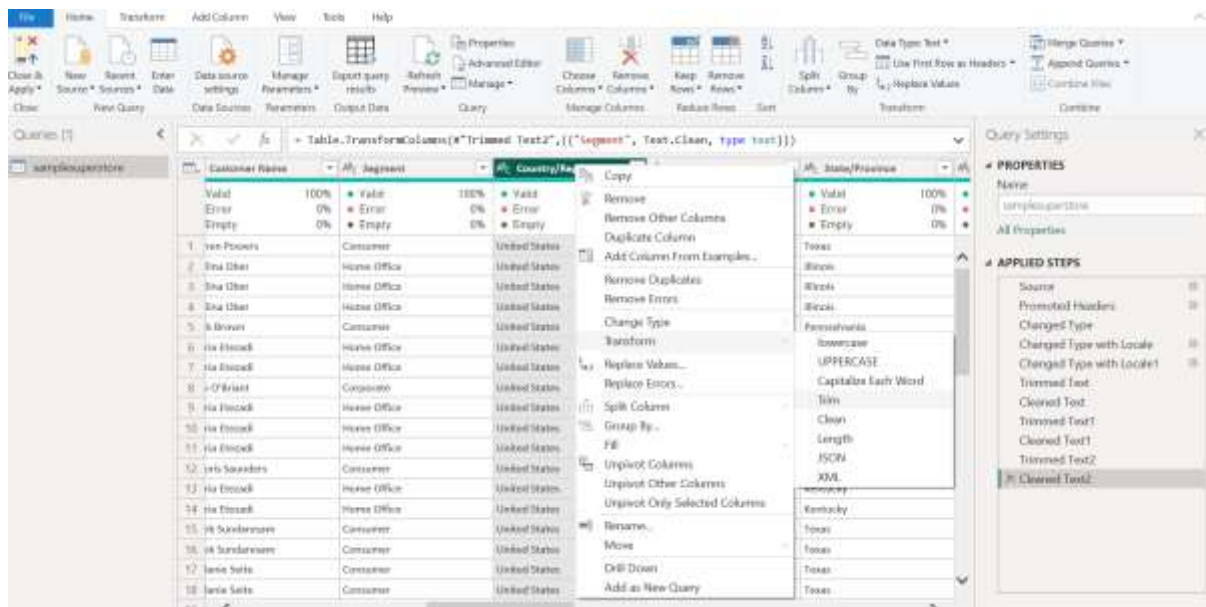
The Transform menu provides various options for cleaning and formatting text-based data. In this example, the Segment column is selected for the text transformation. The Trim function is used to remove unnecessary spaces from the Segment values. This helps maintain clean and consistent data for accurate analysis and reporting.



Applying the Trim Transformation to Segment Data

Trimming Country/Region Data

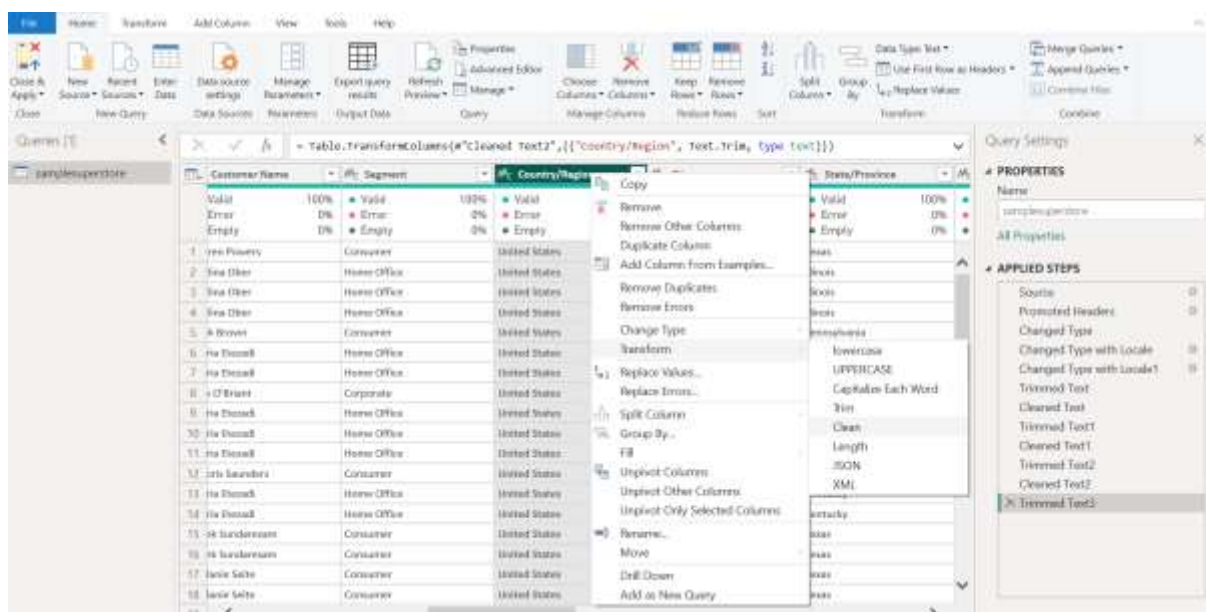
The Country/Region column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from the country and region values. This helps maintain uniform and clean entries throughout the dataset. Clean data improves the accuracy and reliability of farther analysis and reporting.



Applying the Trim Transformation to Country/Region Data

Cleaning Country/Region Data in Power Query

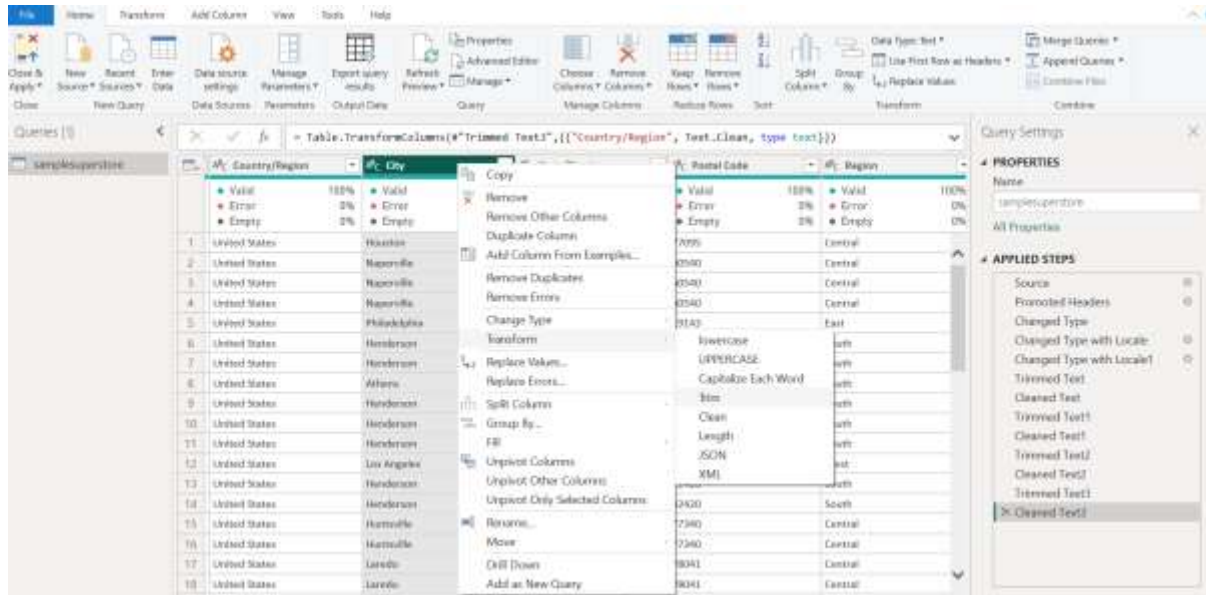
The Transform menu provides options to clean and standardize text-based data. In this example, the Country/Region column is selected for text transformation. The Clean functions help remove unnecessary spaces and unwanted characters. This ensures that country and region values remain consistent for accurate analysis and reporting.



Applying Text Cleaning to Country/Region Data

Trimming City Data in Power Query

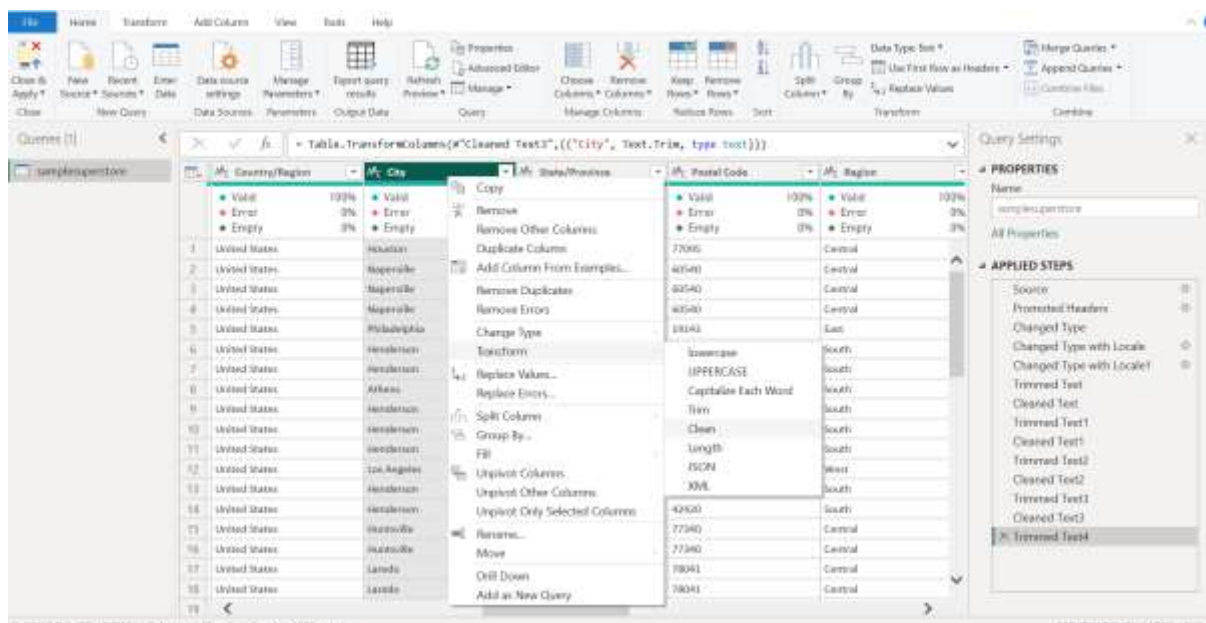
The City column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from city names. This helps maintain consistent and properly formatted city values in the dataset. Clean city data improves the accuracy of analysis and reporting.



Applying the Trim Transformation to City Data

Cleaning City Data

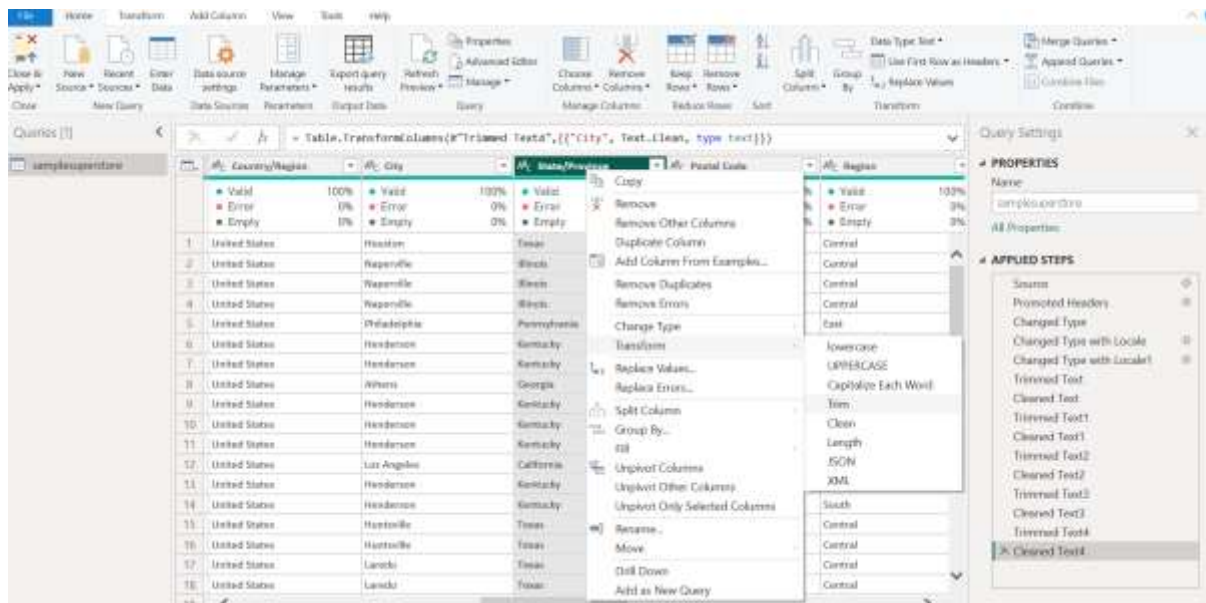
The City column is selected for text cleaning in Power Query. The Clean function removes unwanted or non-printable characters from the data. This helps ensure that country and region values remain clear and consistent. Clean and standardized data supports accurate analysis and reliable reporting.



Applying the Clean Transformation to City Data

Trimming State/Province Data in Power Query

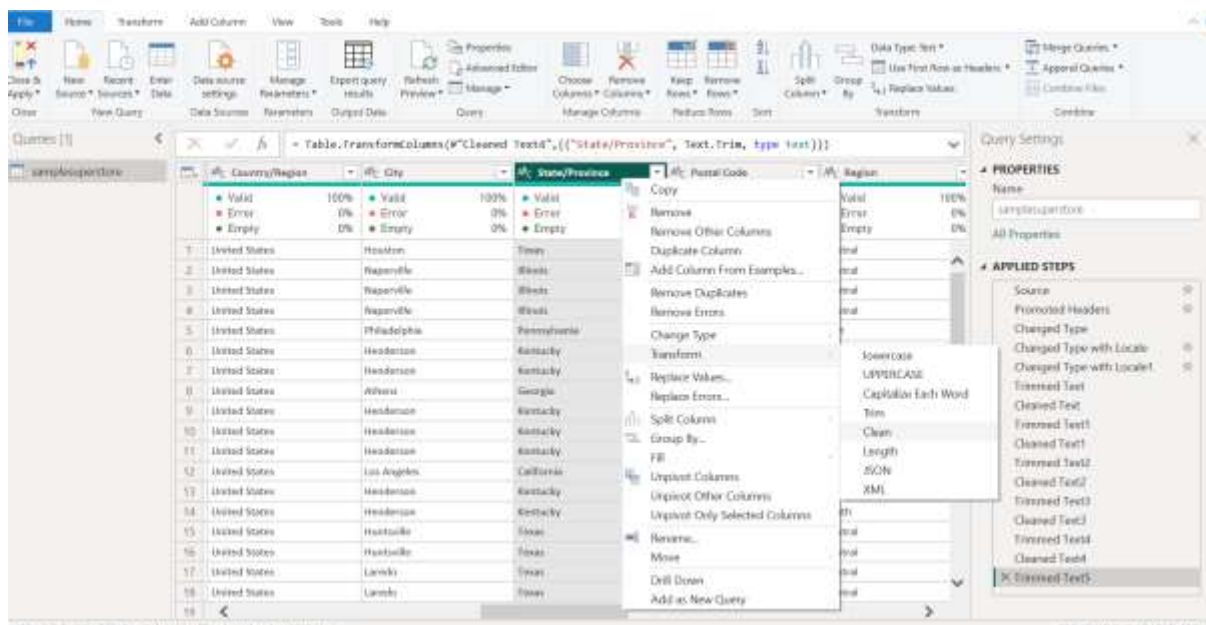
The State/Province column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from state and province names. This helps maintain clean and consistent location values throughout the dataset. Standardized data supports accurate geographical analysis and reporting.



Applying the Trim Transformation to State/Province Data

Cleaning State/Province Data in Power Query

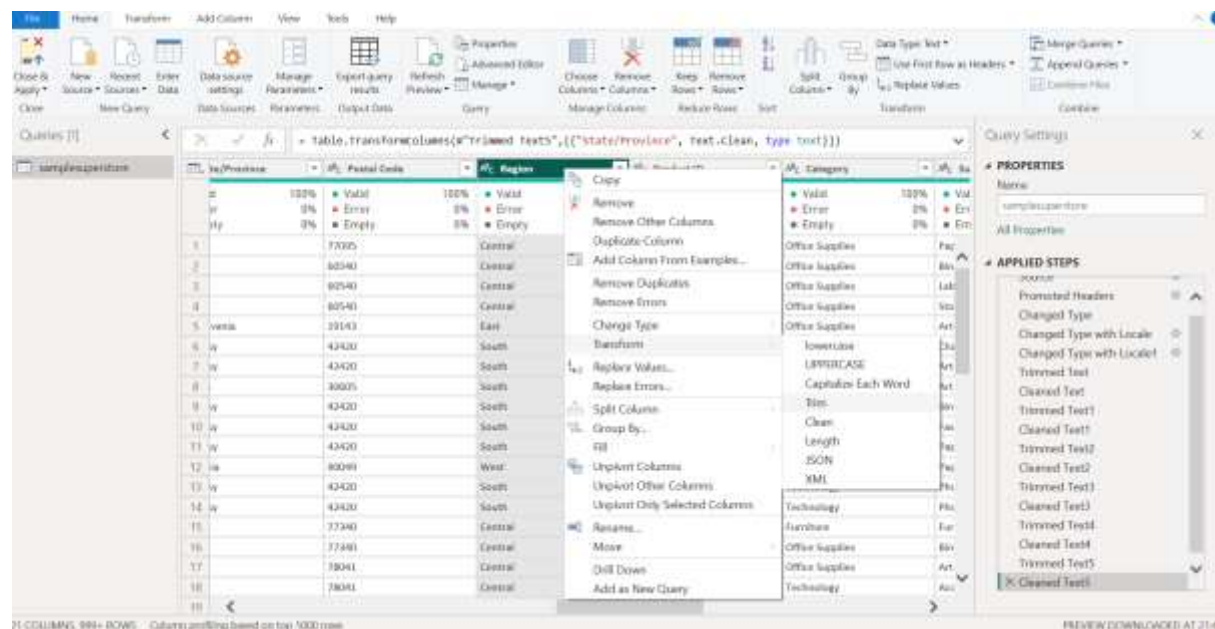
The State/Province column is selected for text transformation in Power Query. The Clean function helps remove unwanted or non-printable characters from state and province names. This ensures that the location values are clean, consistent, and properly formatted. Standardized data improves the accuracy of geographical analysis and reporting.



Applying the Clean Transformation to State/Province Data

Trimming Region Data in Power Query

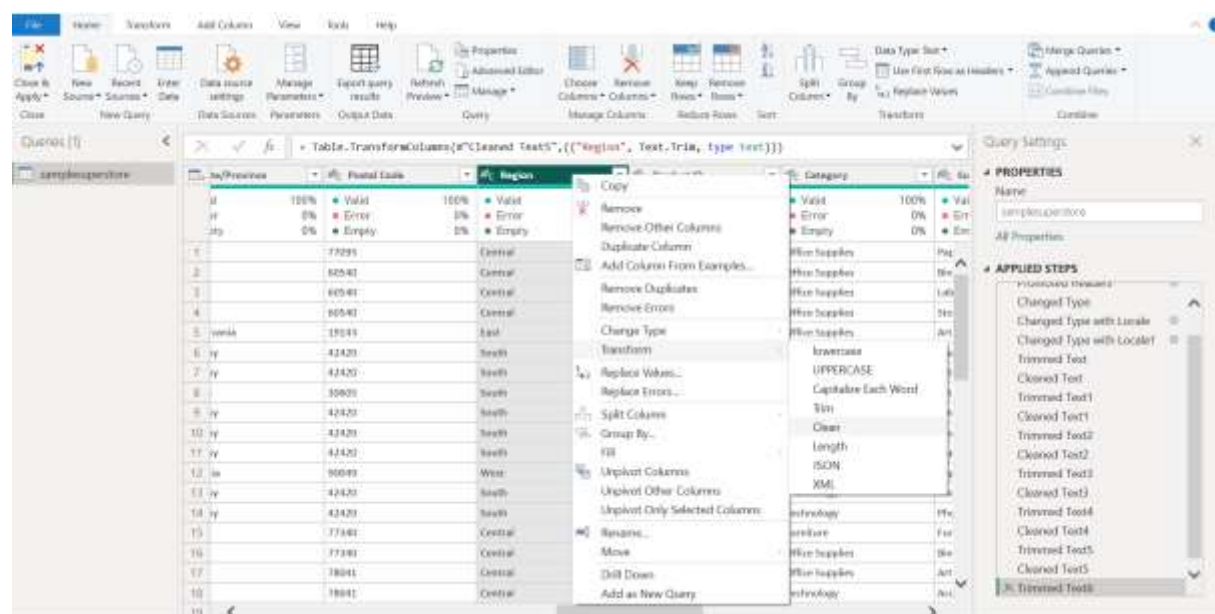
The Region column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from the region values. This helps maintain consistent and properly formatted regional information. Clean and standardized data ensures better accuracy during analysis and reporting.



Applying the Trim Transformation to Region Data

Cleaning Region Data in Power Query

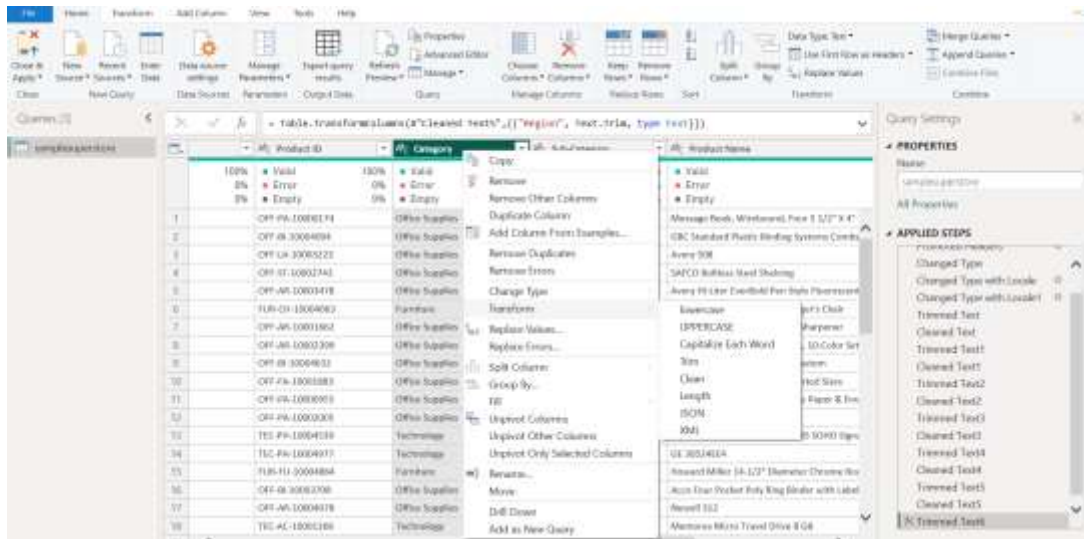
The Region column is selected for text transformation in Power Query. The Clean function helps remove unwanted or non-printable characters from region values. This ensures that regional entries are clear, consistent, and properly formatted. Clean and standardized region data supports accurate analysis and reporting.



Applying the Clean Transformation to Region Data

Trimming Product Category Data in Power Query

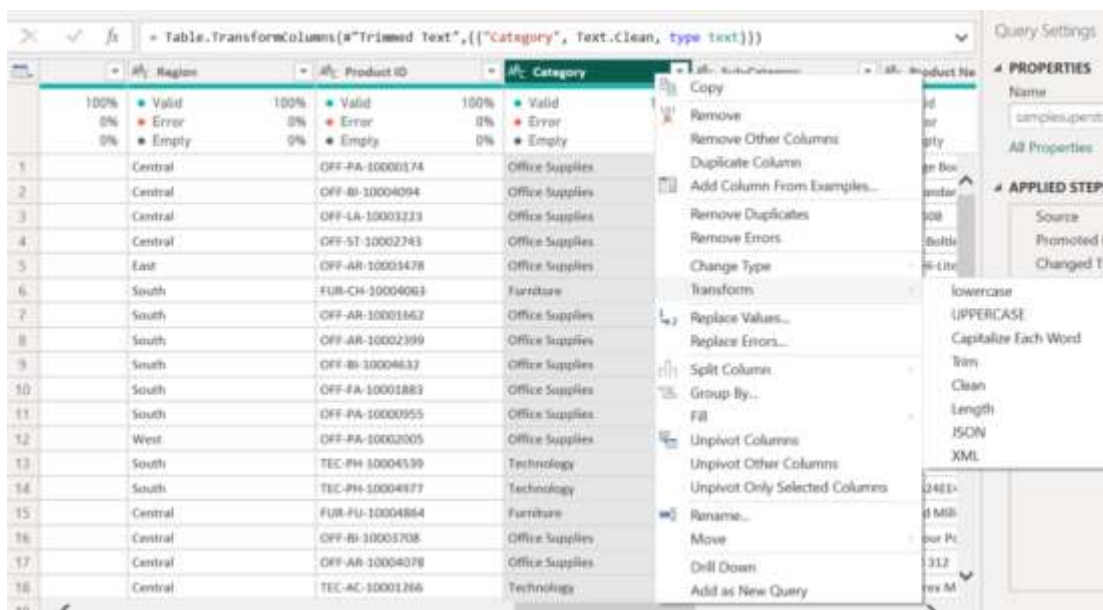
The Transform menu provides various options for trimming and formatting product-related data. In this example, the dataset contains Product ID, Category, Sub-Category, and Product Name columns. Text transformation options such as Trim, Clean, Uppercase, and Lowercase can be applied to standardize the values. These transformations help maintain consistent product information for accurate analysis and reporting.



Applying Text Transformation Options to Product Data in Power Query

Cleaning Category Data in Power Query

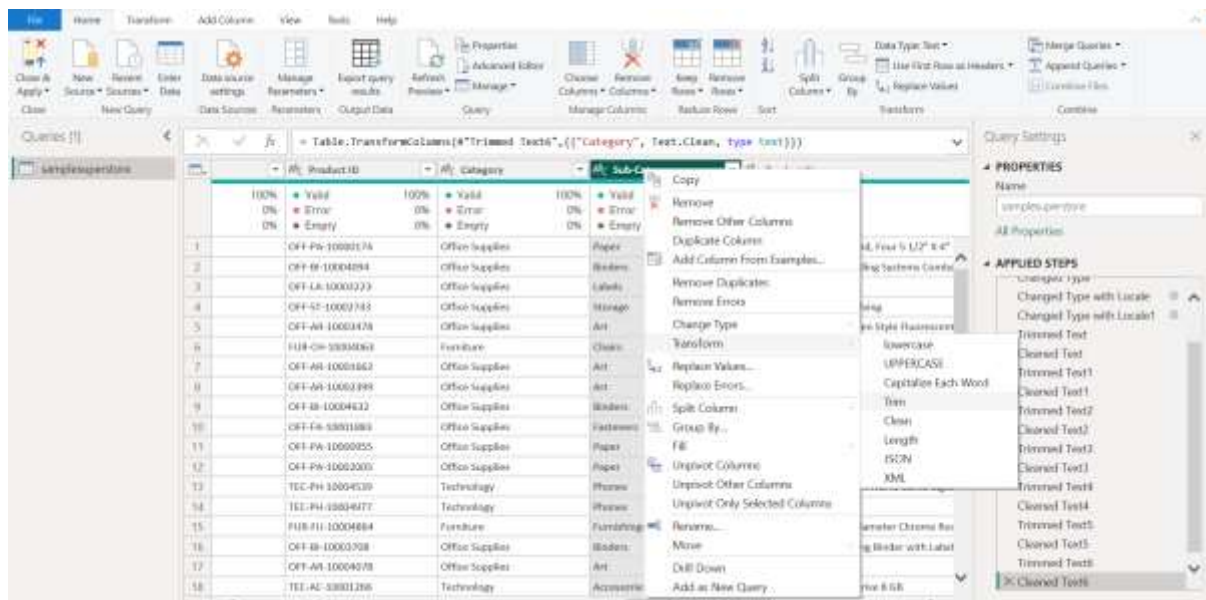
The Category column is selected for text cleaning in Power Query. The Clean function helps remove unwanted or non-printable characters from category values. This ensures that categories such as Office Supplies, Furniture, and Technology remain consistent. Clean and standardized category data supports accurate analysis and reporting.



Applying the Clean Transformation to Category Data in Power Query

Trimming Sub-Category Data in Power Query

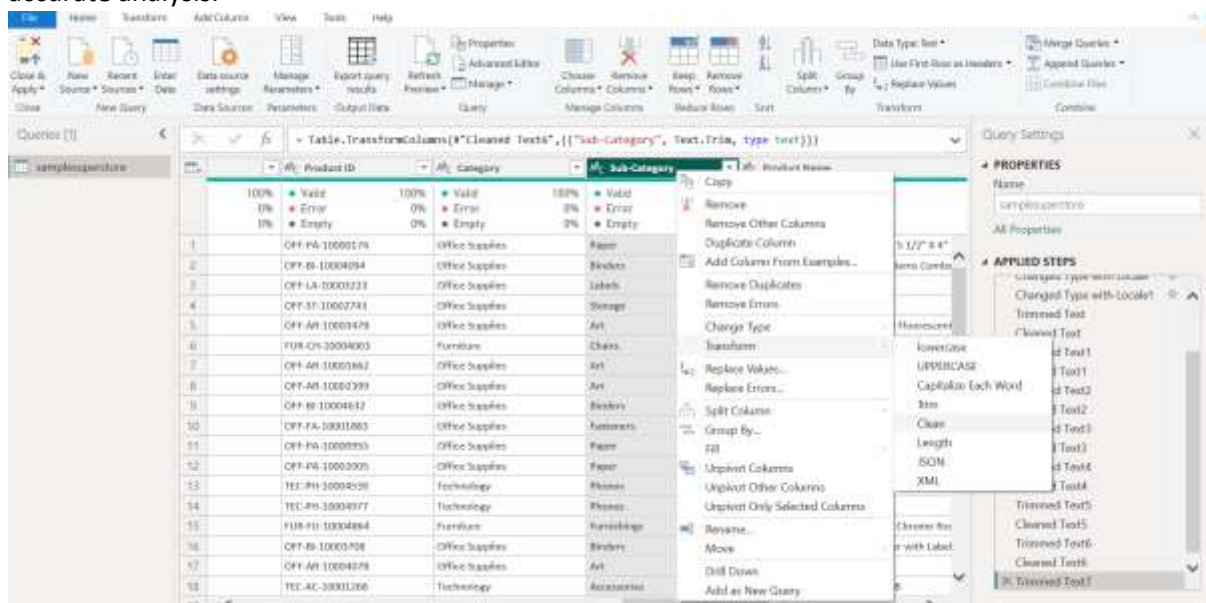
The Sub-Category column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from sub-category values. This helps maintain clean and consistent entries such as Paper, Binders, Labels, and Storage. Standardized sub-category data improves the accuracy of analysis and reporting.



Applying the Trim Transformation to Sub-Category Data in Power Query

Cleaning Product Name Data in Power Query

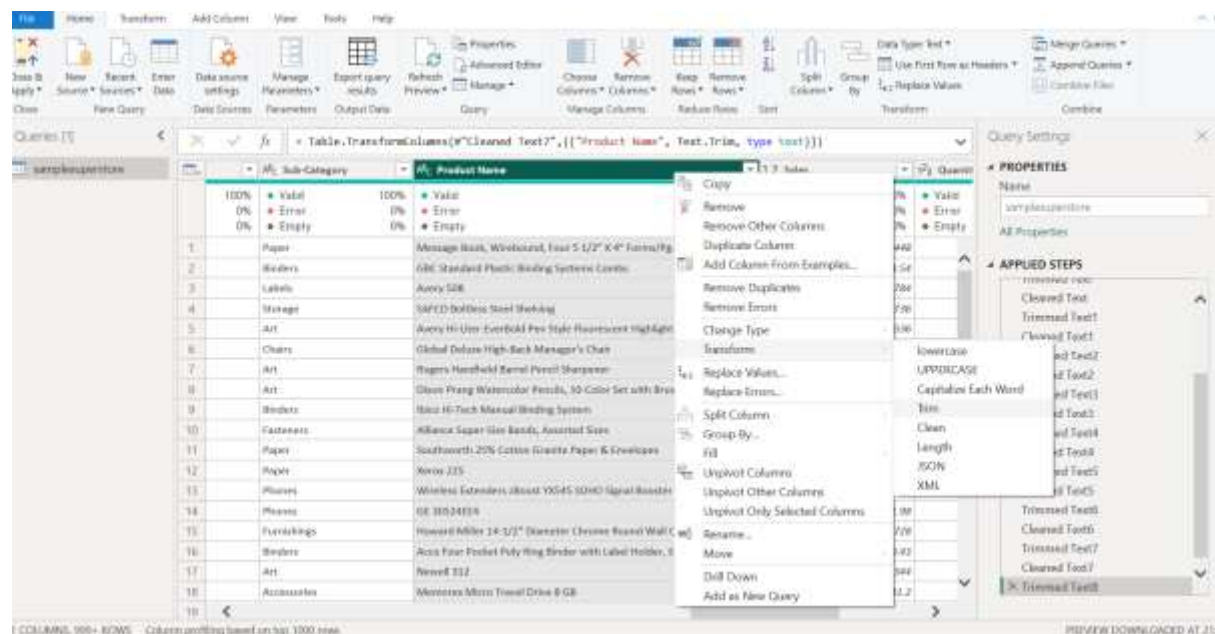
The Product Name column is selected for text transformation in Power Query. The Clean function removes unnecessary spaces from product names. This helps maintain clean and consistent product descriptions across the dataset. Standardized product names improve data quality and support accurate analysis.



Applying the Clean Transformation to Product Name Data in Power

Trimming Product Name Data in Power Query

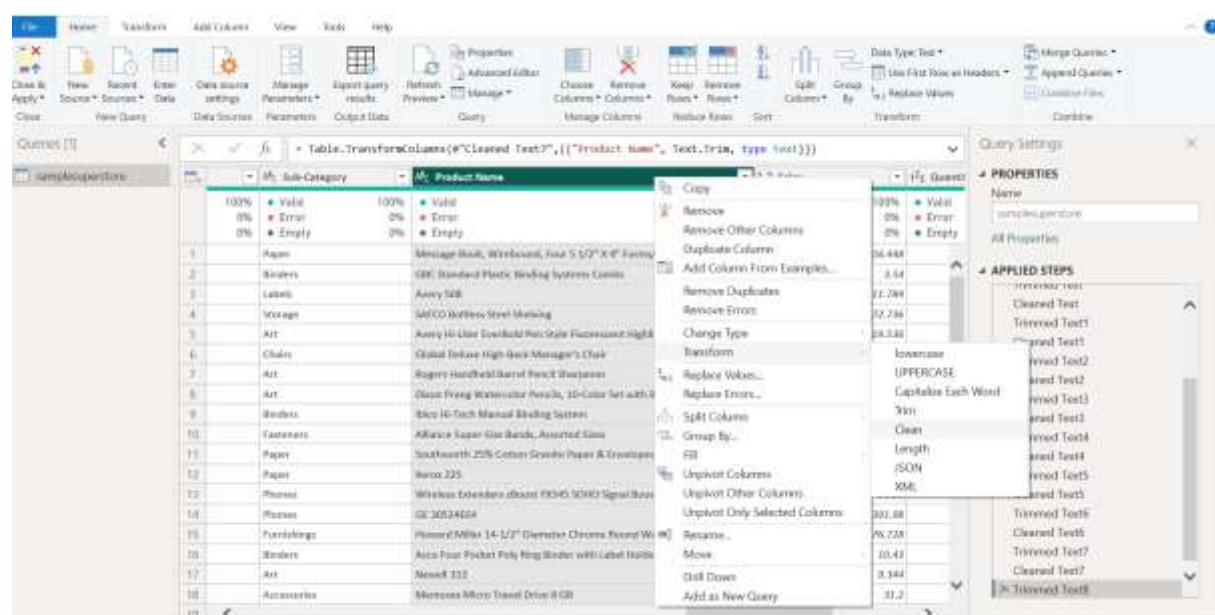
The Product Name column is selected for text transformation in Power Query. The Trim function removes unnecessary spaces from product names. This helps keep product descriptions clean, consistent, and properly formatted. Standardized product names improve data quality and support accurate analysis.



Applying the Trim Transformation to Product Name Data in Power Query

Cleaning Product Name Data in Power Query

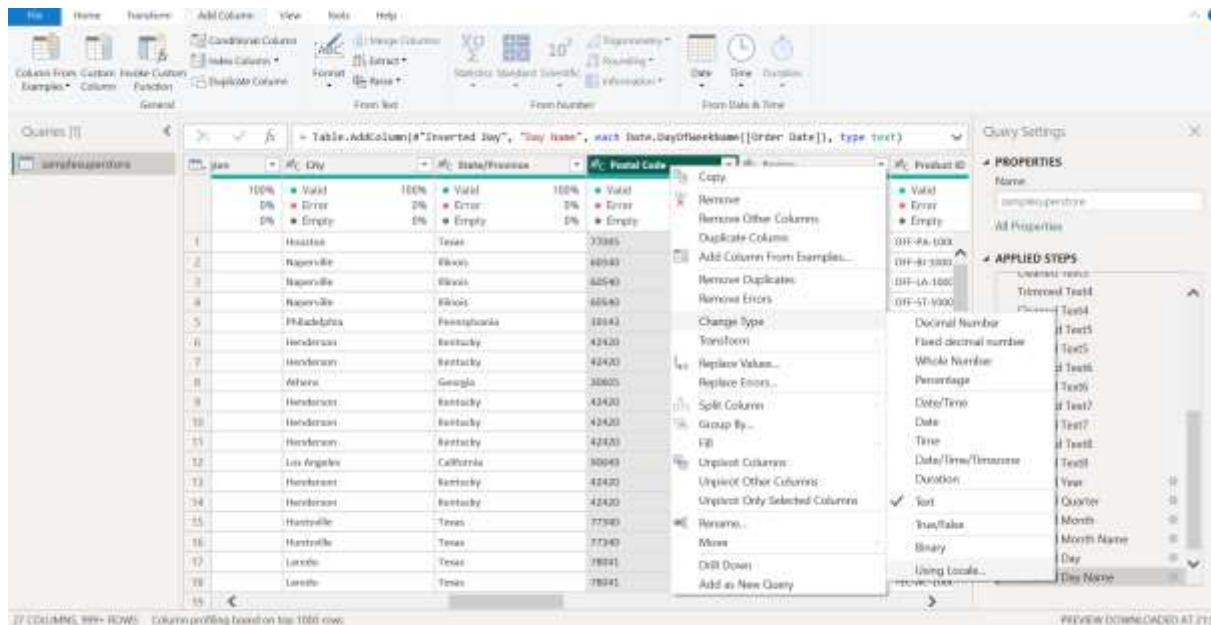
The Product Name column is selected for text transformation in Power Query. The Clean function helps remove unwanted or non-printable characters from product names. This ensures that product descriptions remain clear, consistent, and properly formatted. Clean product data improves data quality and supports accurate analysis and reporting.



Applying the Clean Transformation to Product Name Data in Power Query

Adding a Day Name Column in Power Query

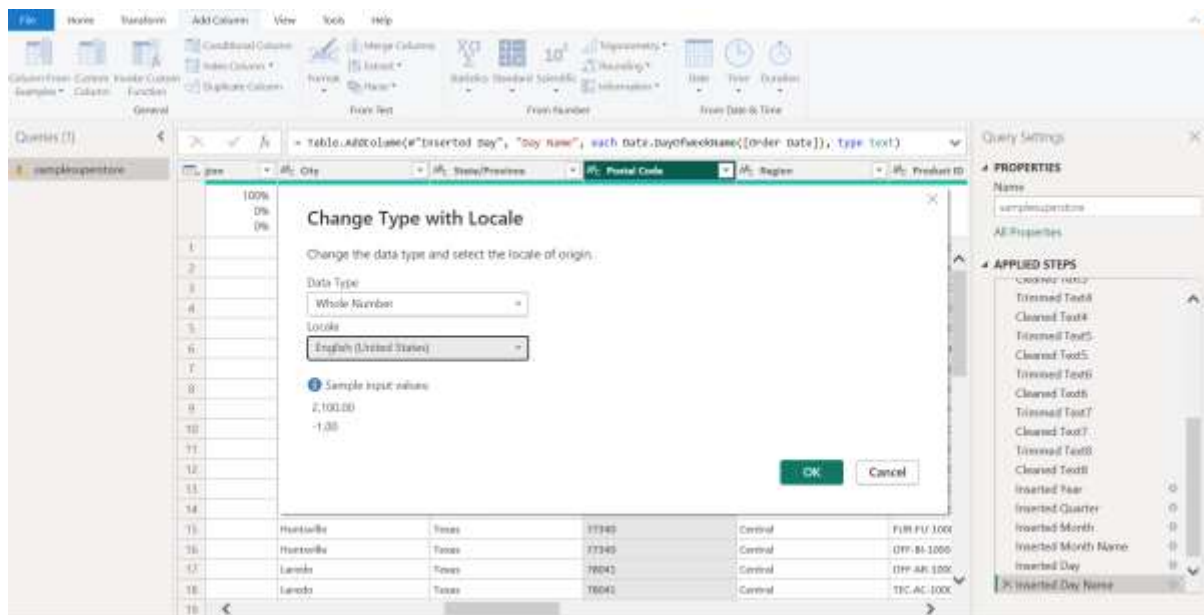
The Add Column feature allows new columns to be created from existing data. In this example, a Day Name column is added using the Order Date column. The formula extracts the day of the week, such as Monday, Tuesday, or Wednesday. This additional column makes it easier to perform day-wise analysis and reporting.



Adding a Day Name Column Based on the Order Date

Changing Data Type with Locale in Power Query

The Change Type with Locale option allows users to convert data based on a specific regional format. In this example, the Whole Number data type is selected with English (United States) as the locale. This ensures that numerical values are interpreted correctly according to the selected regional settings. Using the appropriate locale helps prevent data conversion errors and improves data accuracy.



Changing Data Type and Locale Settings in Power Query

Conclusion

The Superstore Sales dataset was successfully imported and prepared using Power Query Editor. Various cleaning and trimming operations were applied to improve the consistency and quality of the text fields, while suitable data types were assigned to the required columns. The Order Date field was converted into the appropriate date format using the selected locale settings. Additional date attributes such as Year, Quarter, Month, and Day were derived from the Order Date column for better analysis. The Applied Steps panel helped in tracking and verifying the sequence of transformations performed during the preparation process. Overall, the dataset was cleaned, structured, and organized effectively, making it ready for further analysis and visualization in Power BI.